

Direct oral anticoagulants vs warfarin in Asian vs non-Asian patients with atrial fibrillation: a patient-level meta-analysis from COMBINE AF

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Abstract

Background and Aims	Large-scale randomized data comparing clinical outcomes of atrial fibrillation patients of Asian vs non-Asian races are limited.
Methods	Data from A Collaboration Between Multiple Institutions to Better Investigate Non-Vitamin K Antagonist Oral Anticoagulant Use in Atrial Fibrillation, which pooled patient-level data from the four pivotal randomized trials of direct oral anticoagulants (DOACs) vs warfarin in patients with atrial fibrillation, were analysed. Baseline characteristics and clinical outcomes in patients of Asian race (Asians) vs non-Asian race were compared. The relative efficacy and safety of DOACs compared with warfarin in Asians vs non-Asians, assessing for interactions between race and treatment effect, were investigated. Outcomes across the range of body weight and creatinine clearance in Asians were also explored.
Results	A total of 10 212 Asian patients and 61 471 non-Asians were identified. Compared with non-Asians, Asians were on average 3.2 years younger and 20 kg lighter, had worse renal function (mean creatinine clearance

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64.9 vs 77.3 mL/min), and had higher rates of prior stroke/transient ischaemic attack (37.2% vs 26.6%) ($P < .001$ for each). In the warfarin arm (median time in therapeutic range 57.7% for Asians vs 66.2% for non-Asians, $P < .001$), Asians had a higher adjusted risk of stroke/systemic embolic events (SEEs), major bleeding, intracranial haemorrhage, gastrointestinal bleeding, and primary net clinical outcome (stroke/SEE, major bleeding, or death). Compared with warfarin, standard-dose (SD) DOACs significantly reduced the risks in Asians to a greater degree than non-Asians for stroke/SEE (hazard ratio [HR] .65, 95% confidence interval .53–.80 vs HR .86, 95% confidence interval .78–.95), major bleeding (HR .62 [.52–.75] vs HR .91 [.84–.98]), and primary net clinical outcome (HR .76 [.68–.85] vs HR .94 [.90–.98]; $P_{\text{int}} < .02$ for each). Standard-dose DOACs increased gastrointestinal bleeding only in non-Asians (Asians HR .92 [.69–1.23] vs non-Asians HR 1.41 [1.25–1.58], $P_{\text{int}} = .009$). In Asians, SD DOACs reduced the risks of clinical events across the wide range of body weight and creatinine clearance. Compared with SD DOACs, lower-dose DOACs significantly increased the risk of stroke/SEE (HR 1.57 [1.15–2.13]) and secondary net clinical outcome (stroke/SEE, intracranial haemorrhage, or death; HR 1.23 [1.03–1.48]) in Asians.

Conclusions

Clinical outcomes with SD DOACs vs warfarin in patients with atrial fibrillation were generally even more favourable in patients of Asian than non-Asian race. Compared with warfarin, SD DOACs did not increase the risk of gastrointestinal bleeding in Asians. Standard-dose DOACs are preferred over warfarin or lower-dose DOACs in patients of Asian race.

Structured Graphical Abstract

Key Question

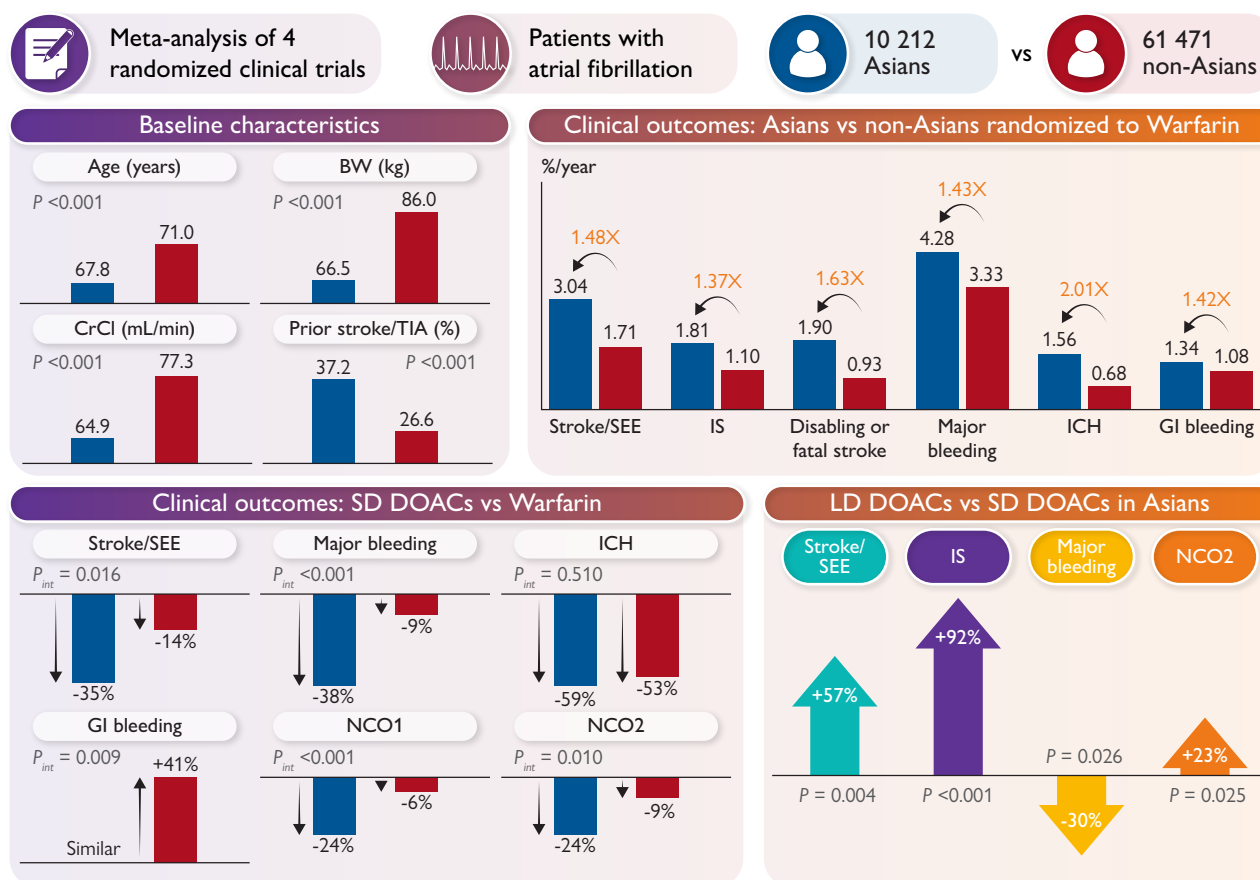
Do clinical outcomes and treatment effects of direct oral anticoagulants (DOACs) vs warfarin differ between patients of Asian vs non-Asian race with atrial fibrillation (AF)?

Key Finding

In this meta-analysis of 4 trials, Asians were younger, had lower weight, poorer renal function, and had a higher risk of stroke than non-Asians. Compared with warfarin, standard-dose DOACs resulted in greater reductions in stroke/systemic embolic events, major bleeding, and net clinical outcomes in Asians vs non-Asians.

Take Home Message

Standard-dose DOACs should be preferred over warfarin for embolic event prevention in Asian AF patients.



Using patient-level data from four pivotal randomized trials in the COMBINE AF dataset, we compared clinical characteristics and outcomes between Asian and non-Asian patients with atrial fibrillation. Asians had worse outcomes on warfarin, whereas standard-dose DOACs provided greater reductions in stroke/SEE, major bleeding, and net clinical outcomes without excess gastrointestinal bleeding. Standard-dose DOACs should be preferred over warfarin or lower-dose DOACs for stroke prevention in Asian patients with atrial fibrillation.

NCO1 - composite of stroke, SEE, major bleeding or all-cause death; NCO2 - composite of stroke, SEE, ICH or all-cause death

BW, body weight; CrCl, creatinine clearance; DOACs, direct oral anticoagulants; GI, gastrointestinal; ICH, intracranial hemorrhage; IS, ischemic stroke; LD DOACs, lower-dose direct oral anticoagulants; NCO, net clinical outcome; SD DOACs, standard-dose direct oral anticoagulants; SEE, systemic embolic event; TIA, transient ischemic attack

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Keywords

Atrial fibrillation • Warfarin • DOACs • Asian race • COMBINE AF

Introduction

Clinical trials provide the most important base for clinical practice guidelines and patient care. Global enrolment allows evaluation of the results to different regions, races, and ethnicities. Although Asia has an enormous population, Asian participants (and authors) have been under-represented in cardiovascular clinical trials.^{1,2} Among cardiometabolic trials published in the prior decade, only 8.3% participants identified as being of Asian race and 7.7% of total enrolment occurred in Asia-Pacific.¹ This issue is particularly important for atrial fibrillation (AF) considering that Asia has the largest burden of AF in the world.^{3,4} It was estimated that by 2050, AF will be diagnosed at least in 72 million individuals in Asia, with $\approx$ 3 million AF-related strokes.⁵

Differences in the risks of ischaemic and bleeding events in patients of Asian vs non-Asian races have been reported.^{6,7} Prior randomized trials in AF comparing warfarin with direct oral anticoagulants (DOACs) demonstrated significant differences regarding clinical characteristics and risks of clinical events between Asian and non-Asian race patients with AF.^{8–11} Generally, Asian AF patients were younger, had a lower body weight (BW), a lower creatinine clearance (CrCl), and a higher risk of stroke and intracranial haemorrhage (ICH) on warfarin compared with non-Asians. Prior analyses have suggested that treatment effects of DOACs compared with warfarin may be different in patients of Asian vs non-Asian race, but the number of Asians included was modest.^{9–12}

The COMBINE AF (A Collaboration Between Multiple Institutions to Better Investigate Non-Vitamin K Antagonist Oral Anticoagulant Use in Atrial Fibrillation) database contains individual patient data from the four pivotal trials of DOACs vs warfarin in patients with AF.¹³ Patient-level meta-analyses offer a number of advantages, including the ability to evaluate small subgroups, infrequent outcomes, and continuous variables and to use multivariate adjustment, time-to-event censored survival data, and application of consistent follow-up time across trials. The COMBINE AF database not only provides a large sample size to study the differences between Asian and non-Asian AF patients but also allows for analyses of continuous variables (e.g. BW and CrCl) and a more thorough assessment of treatment effect heterogeneity by subgroup.

The goal of this analysis was to investigate the baseline characteristics and to compare risks of thromboembolic and bleeding events of patients of Asian race, regardless of country of enrolment, to those in non-Asians in the COMBINE AF data. We also aimed to study the efficacy and safety of DOACs compared with warfarin for Asians and non-Asians tested for interactions and across the range of BW and CrCl in Asians.

Methods

COMBINE AF dataset

The design and rationale of COMBINE AF have been described previously.^{13,14} Briefly, COMBINE AF incorporated individual patient data from 71 683 deidentified patients from four pivotal randomized clinical trials comparing DOACs with warfarin in patients with AF, including Randomized Evaluation of Long-term Anticoagulation Therapy (RE-LY), Rivaroxaban Once Daily Oral Direct Factor Xa Inhibition Compared with Vitamin K Antagonism

for Prevention of Stroke and Embolism Trial in Atrial Fibrillation (ROCKET AF), Apixaban for Reduction in Stroke and Other Thromboembolic Events in Atrial Fibrillation (ARISTOTLE), and Effective Anticoagulation With Factor Xa Next Generation in Atrial Fibrillation–Thrombolysis in Myocardial Infarction 48 (ENGAGE AF-TIMI 48), who were randomized to warfarin or DOACs.

For these analyses, standard-dose (SD) DOACs were defined as dabigatran 150 mg twice daily, rivaroxaban 20 mg (or 15 mg if dose reduction criteria were met) once daily, apixaban 5 mg (dose reduced to 2.5 mg) twice daily, or edoxaban 60 mg (dose reduced to 30 mg) once daily. Lower-dose (LD) DOACs were defined as dabigatran 110 mg twice daily or edoxaban 30 mg (dose reduced to 15 mg) once daily.

Study population

Among 71 683 patients, there were 10 212 (14.2%) patients of Asian race and 61 471 (85.8%) of non-Asian race. Race and ethnicity were self-identified, regardless of country of enrolment, into one of six mutually exclusive categories (Asian, Caucasian, Black, American Indian/Alaskan Native, Native Hawaiian, and others). Among 61 471 patients of non-Asian race, a total of 25 129 patients were randomized to warfarin, 11 116 to LD DOACs, and 25 226 to SD DOACs. Among 10 212 patients of Asian race, a total of 4143 patients were randomized to warfarin, 1933 to LD DOACs, and 4136 to SD DOACs (see [Supplementary data online, Figure S1](#)). The distributions of DOAC dosing of rivaroxaban, apixaban, and edoxaban for Asians and non-Asians are shown in [Supplementary data online, Figure S2](#). Overall, the proportion of patients who met the dose reduction criteria for reduced-dose DOACs were higher for Asians compared with non-Asians ($P < .001$).

Clinical endpoints

Outcome definitions in COMBINE AF have been described previously.¹³ All outcomes were adjudicated as described in the individual trials and were assessed as time to first event. Efficacy endpoints included stroke or systemic embolic events (SEEs), ischaemic stroke, disabling or fatal stroke, myocardial infarction, all-cause death, and cardiovascular mortality. Safety outcomes included major bleeding (MB), ICH, fatal bleeding, major or clinically relevant non-major (CRNM) bleeding, and major gastrointestinal (GI) bleeding. Two net clinical outcomes (NCOs) were analysed: (i) NCO1, composite of stroke, SEE, MB, or all-cause death, and (ii) NCO2, composite of stroke, SEE, ICH, or all-cause death.

For efficacy outcomes, the intention-to-treat population was used. To account for different follow-up durations across trials and to set a comparable follow-up duration, events were censored at 32 months. For safety and net outcomes, the safety population was used as defined by each of the individual trials but typically included participants who received at least one dose of a study drug and were followed up for events occurring between the date the participant began treatment with the study drug and up to 2 days after the participant discontinued study drug, as described previously.^{13–16}

Statistical analysis

Baseline characteristics between patients of Asian and non-Asian race were compared, and candidate variables were identified to include in subsequent adjusted analyses. Data were presented as the mean value and standard deviation for continuous variables and proportions for categorical variables. Differences in clinical characteristics were evaluated using Pearson's χ^2 test for categorical variables and the Kruskal–Wallis test for continuous variables. The time in therapeutic range (TTR), defined as an international normalized ratio (INR) between 2 and 3 in the warfarin group, was calculated by means of linear interpolation.¹⁷

The risks (expressed as adjusted hazard ratios [aHRs]) of clinical events for Asians vs non-Asians treated with warfarin were compared using a stratified Cox proportional hazards model stratified by trial, adjusted for age, sex, BW, pattern of AF, systolic blood pressure, diastolic blood pressure, CrCl, history of heart failure, hypertension, diabetes mellitus, stroke/transient ischaemic attack, myocardial infarction, peripheral arterial diseases, and prior use of vitamin K antagonist. The proportional hazards assumption was tested using the Schoenfeld residual test, which showed no non-proportionality. Since a TTR > 70% is considered a high level of effective anticoagulation, with lower TTRs considered a reason to switch from warfarin to a DOAC in current guidelines,^{18,19} we performed a sensitivity analysis comparing outcomes between Asians and non-Asians in patients with TTR > 70% and ≤70%.

The effect modification of race (Asian vs non-Asian) on clinical outcomes with SD DOACs vs warfarin was evaluated by including an interaction term between Asian race and randomized treatment arm. In addition, outcomes in patients who received SD DOAC vs warfarin stratified by the presence or absence of dose-reduction criteria were determined. However, since the three trials with factor Xa inhibitors (FXai; rivaroxaban, apixaban and edoxaban) used different criteria for dose reduction (and there was no dose reduction of dabigatran in RE-LY), we compared SD FXai with warfarin stratified by criteria available in two of the three trials, namely, 'BW ≤ 60 kg or CrCl ≤ 50 mL/min' and 'BW > 60 kg and CrCl > 50 mL/min'. The former subgroup represents two factors identifying patients at high risk for thromboembolic and bleeding events.^{16,20–25}

Among patients with Asian race, we evaluated treatment effects of SD DOACs vs warfarin across continuous CrCl and BW values. Restricted cubic splines were constructed using three to five knots to display the event probabilities by treatment allocation for clinical outcomes across continuous BW and CrCl allowing for nonlinearity. A $P < .05$ was considered statistically significant, with no adjustment for multiple comparisons. All analyses were performed with SAS version 9.4.

Results

Clinical characteristics of patients with Asian and non-Asian race

The baseline characteristics of patients of Asian vs non-Asian races are shown in [Table 1](#). Clinical characteristics of Asians and non-Asians randomized to warfarin, SD DOACs, and LD DOACs are shown in [Supplementary data online, Tables S1–S3](#). Compared with non-Asians, Asians were on average 3.2 years younger (mean age 67.8 years vs 71.0 years, $P < .001$), 19.5 kg lighter (mean BW 66.5 vs 86.0 kg, $P < .001$), and more likely male (64.1% vs 62.5%, $P = .002$) with a lower CrCl (mean 64.9 vs 77.3 mL/min, $P < .001$). Nearly twice as many Asians than non-Asians met dose reduction criteria for DOACs (31.0% vs 16.0%, $P < .001$) largely due to their lower BW and CrCl. The histograms about the distributions of BW and CrCl for Asians and non-Asians are shown in [Supplementary data online, Figure S3](#). The CHA₂DS₂-VASC (Congestive heart failure, Hypertension, Age ≥ 75 years, Diabetes, Stroke/transient ischaemic attack, Vascular disease, Age 65 to 74 years, Sex category) score was lower for Asians compared with non-Asians (mean score 3.7 vs 4.0, $P < .001$). Lower rates of prior congestive heart failure, hypertension, diabetes mellitus, and myocardial infarction, but higher rates of prior stroke/transient ischaemic attack (37.2% vs 26.6%) were present in the Asian population. There was no significant difference in the rate of premature

permanent discontinuation of trial drugs between Asians (27.9%) and non-Asians (28.7%) ($P = .11$).

Time in therapeutic range with warfarin

The median TTR was on average 8.5% lower in Asian patients treated with warfarin than in non-Asians (median [25th–75th percentiles] 57.7% [41.7–70.1] vs 66.2% [52.7–76.8], $P < .001$). Only 25.2% of Asians had a TTR above 70% compared with 41.1% of non-Asians ($P < .001$).

Risks of clinical events in Asians vs non-Asians with warfarin

When compared with non-Asians, the risks of stroke/SEE (aHR 1.48, 95% confidence interval [CI] 1.25–1.75, $P < .001$), ischaemic stroke (aHR 1.37 [1.11–1.70], $P = .004$), disabling or fatal stroke (aHR 1.63 [1.31–2.02], $P < .001$), and major adverse cardiovascular events (MACEs) (aHR 1.16 [1.03–1.30], $P = .013$) were higher for Asians ([Figure 1](#)). Asians treated with warfarin had higher risk of MB (aHR 1.43 [1.24–1.66], $P < .001$), ICH (aHR 2.01 [1.55–2.61], $P < .001$), major or CRNM bleeding (aHR 1.53 [1.40–1.67], $P < .001$) and GI bleeding (aHR 1.42 [1.10–1.83], $P = .006$). The risk of NCO1 was higher in Asians than non-Asians (aHR 1.15 [1.05–1.26], $P = .004$), while there were no significant differences between the groups in the adjusted risks for myocardial infarction, mortality endpoints, or other composite endpoints in the warfarin arm ([Figure 1](#)). The outcomes in Asians compared with non-Asians were generally consistent among patients with a TTR > 70% (Asians, $n = 1013$; non-Asians, $n = 10\,088$) or ≤70% (Asians, $n = 3007$; non-Asians, $n = 14\,431$), P -value for interaction (P_{int}) > .05 for all endpoints except for disabling or fatal stroke (Asians vs non-Asians: aHR 2.43 in TTR > 70% and 1.35 in TTR ≤ 70%, $P_{\text{int}} = .017$) (see [Supplementary data online, Figure S4](#)). Compared with a TTR ≤ 70%, a TTR > 70% was associated with a lower risk of disabling or fatal stroke in non-Asians (.63%/year vs 1.12%/year, $P < .001$), but not in Asians (1.76%/year vs 1.85%/year, $P = .792$).

Efficacy, safety, and net clinical outcomes of patients treated with standard-dose direct oral anticoagulants compared with warfarin for Asians and non-Asians

Clinical characteristics of Asians and non-Asians randomized to DOACs and warfarin in the four trials combined and within each trial are shown in [Supplementary data online, Tables S4–S8](#). Baseline characteristics were generally balanced between randomized treatment groups. Standard-dose DOACs, as compared to warfarin, reduced the risks of stroke/SEE (hazard ratio [HR] .65 vs .86, $P_{\text{int}} = .016$), disabling or fatal stroke (HR .63 vs .88, $P_{\text{int}} = .019$), all-cause death/stroke/SEE (HR .77 vs .92, $P_{\text{int}} = .011$), and MACE (HR .76 vs .91, $P_{\text{int}} = .019$) to a greater degree in Asians compared with non-Asians ([Figure 2](#)).

Among Asians, there was a significant 38% relative reduction in MB with SD DOACs compared with warfarin (2.64%/year vs 4.28%/year, HR .62 [.52–.75], $P < .001$), while in non-Asians, the relative reduction was a more modest 9% (3.02%/year vs 3.33%/year; HR .91 [.84–.98], $P = .014$; $P_{\text{int}} < .001$). The risk of ICH was similarly reduced with SD DOACs compared with warfarin in Asians (HR .41 [.29–.58], $P < .001$) and non-Asians (HR

Table 1 Baseline characteristics of patients of Asian and non-Asian race

Variables	Asians (n = 10 212)	Non-Asians (n = 61 471)	P-value
Age, years, mean (SD)	67.8 (10)	71.0 (9.1)	<.001
Age ≥ 75	2957/10 212 (29)	24 682/61 471 (40.2)	<.001
Age 65–75	4029/10 212 (39.5)	22 782/61 471 (37.1)	<.001
Age < 65	3226/10 212 (31.6)	14 007/61 471 (22.8)	<.001
Female sex	3664/10 212 (35.9)	23 050/61 470 (37.5)	.002
Region			<.001
North America	149/10 212 (1.5)	18 236/61 471 (29.7)	
Latin America	16/10 212 (.2)	8931/61 471 (14.5)	
Europe	104/10 212 (1.0)	34 248/61 471 (55.7)	
Asia	9943/10 212 (97.4)	56/61 471 (.1)	
Body weight, kg, mean (SD)	66.5 (13.3)	86 (19.5)	<.001
<60	3126/10 199 (30.7)	3602/61 410 (5.9)	<.001
60 to <100	6916/10 199 (67.8)	45 047/61 410 (73.4)	<.001
≥100	157/10 199 (1.5)	12 761/61 410 (20.8)	<.001
CrCl, mL/min, mean (SD)	64.9 (23)	77.3 (33.8)	<.001
CrCl ≤ 50 mL/min	2906/10 199 (28.5)	11 117/61 372 (18.1)	<.001
Body weight ≤ 60 kg or CrCl ≤ 50 mL/min	4640/10 212 (45.4)	12 568/61 471 (20.4)	<.001
Paroxysmal atrial fibrillation	1921/10 210 (18.8)	14 688/61 458 (23.9)	<.001
Systolic blood pressure, mmHg, mean (SD)	128.8 (16.3)	131.4 (16.3)	<.001
Diastolic blood pressure, mmHg, mean (SD)	77.8 (10.4)	78.2 (10.3)	<.001
Comorbidities			
Heart failure	4127/10 211 (40.4)	29 149/61 469 (47.4)	<.001
Hypertension	8077/10 212 (79.1)	54 786/61 470 (89.1)	<.001
Diabetes mellitus	3015/10 212 (29.5)	19 072/61 470 (31)	.002
Previous stroke/transient ischaemic attack	3799/10 212 (37.2)	16 348/61 470 (26.6)	<.001
Prior myocardial infarction	772/10 208 (7.6)	9719/61 459 (15.8)	<.001
Peripheral arterial disease	171/10 147 (1.7)	3074/61 310 (5)	<.001
Vascular diseases (myocardial infarction or peripheral arterial disease)	920/10 212 (9)	11 825/61 471 (19.2)	<.001
Prior gastrointestinal bleeding ^a	262/7337 (3.6)	1768/46 229 (3.8)	.291
CHADS ₂ score, mean (SD)	2.5 (1.1)	2.6 (1.2)	<.001
CHA ₂ DS ₂ -VASc score, mean (SD)	3.7 (1.5)	4 (1.5)	<.001
Previous use of vitamin K antagonist	5396/10 212 (52.8)	43 496/61 468 (70.8)	<.001
Medications at randomization			
Aspirin	3880/10 192 (38.1)	20 335/61 440 (33.1)	<.001
Thienopyridine	677/10 192 (6.6)	1496/61 440 (2.4)	<.001
Beta-blockers	4982/10 118 (49.2)	41 109/61 220 (67.1)	<.001
Non-steroidal anti-inflammatory drugs	337/10 192 (3.3)	2612/61 440 (4.3)	<.001
Amiodarone	1036/10 118 (10.2)	6593/61 220 (10.8)	.11
Digoxin or digitalis preparation	3645/10 118 (36)	19 210/61 220 (31.4)	<.001
Proton pump inhibitor	721/10 113 (7.1)	7817/61 197 (12.8)	<.001

Continued

Table 1 Continued

Variables	Asians (n = 10 212)	Non-Asians (n = 61 471)	P-value
Time in therapeutic range ^b , %, median [IQR]	57.7 [41.7–70.1]	66.2 [52.7–76.8]	<.001
Time in therapeutic range > 70%	1013/4020 (25.2)	10 088/24 519 (41.1)	<.001

Data shown are n/N (%) unless otherwise specified.
CrCl, creatinine clearance; IQR, interquartile range; SD, standard deviation.
^aData on prior gastrointestinal bleeding were not available in RE-LY trial.
^bWarfarin arm only.

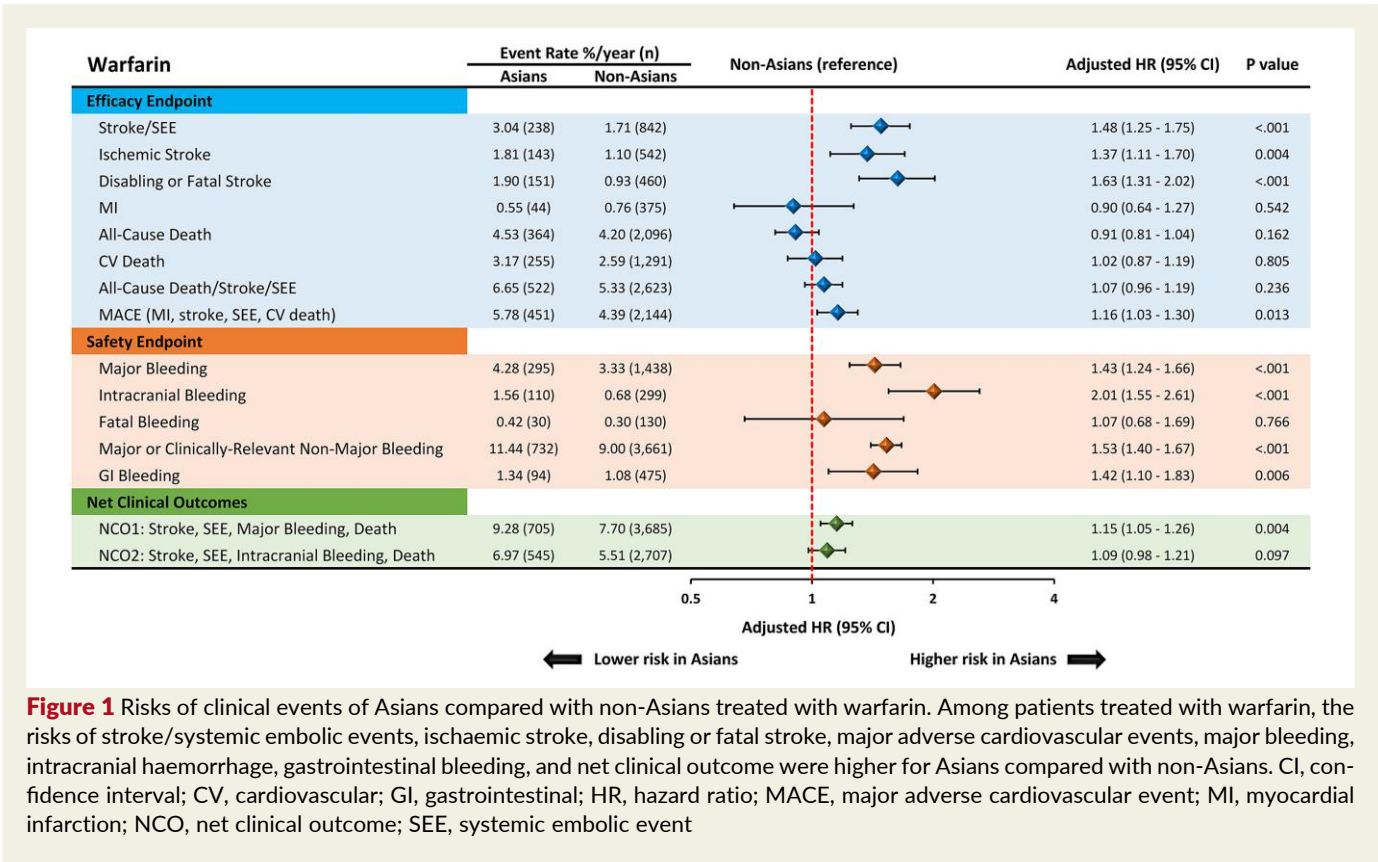


Figure 1 Risks of clinical events of Asians compared with non-Asians treated with warfarin. Among patients treated with warfarin, the risks of stroke/systemic embolic events, ischaemic stroke, disabling or fatal stroke, major adverse cardiovascular events, major bleeding, intracranial haemorrhage, gastrointestinal bleeding, and net clinical outcome were higher for Asians compared with non-Asians. CI, confidence interval; CV, cardiovascular; GI, gastrointestinal; HR, hazard ratio; MACE, major adverse cardiovascular event; MI, myocardial infarction; NCO, net clinical outcome; SEE, systemic embolic event

.47 [.38–.57], $P < .001$; $P_{\text{int}} = .51$). The risk of GI bleeding was significantly higher with SD DOACs compared with warfarin among non-Asians (HR 1.41 [1.25–1.58]), while the risk was similar between SD DOACs and warfarin in Asians (HR .92 [.69–1.23]; $P_{\text{int}} = .009$).

Compared with warfarin, the NCOs were more favourable with SD DOACs in Asians compared with non-Asians. The HRs for the NCO1 were .76 [.68–.85] in Asians vs .94 [.90–.98] in non-Asians ($P_{\text{int}} < .001$), and for the NCO2 were .76 [.67–.86] vs .91 [.86–.96] ($P_{\text{int}} = .010$).

The aforementioned greater advantages of SD DOACs compared with warfarin in Asians as compared with in non-Asians were present particularly among high-risk patients with a $\text{CrCl} \leq 50 \text{ mL/min}$ or $\text{BW} \leq 60 \text{ kg}$ for stroke/SEE, MB, and NCOs (see [Supplementary data online, Figure S5](#)). However, the advantages of SD DOACs over warfarin were not different between races in the complementary subgroup (i.e. patients

with $\text{CrCl} > 50 \text{ mL/min}$ and $\text{BW} > 60 \text{ kg}$) ($P_{\text{int}} > .05$) for each endpoint (see [Supplementary data online, Figure S6](#)).

Several significant treatment interactions were observed within key subgroups of Asian patients (see [Supplementary data online, Figure S7–S9](#)). In particular, even greater benefits with SD DOAC compared with warfarin were seen in Asian patients who had a higher $\text{CHA}_2\text{DS}_2\text{-VASc}$ score (for the endpoints of stroke/SEE, ICH, and NCO2) and in Asians with diabetes (for the endpoints of stroke/SEE and disabling or fatal stroke).

Standard-dose direct oral anticoagulants vs warfarin in Asians across body weight and creatinine clearance

[Figures 3](#) and [4](#) show the event probabilities of SD DOACs and warfarin across BW (ranging from 40 to 110 kg) and CrCl (ranging from 30 to 140 mL/min), respectively, among Asians. In

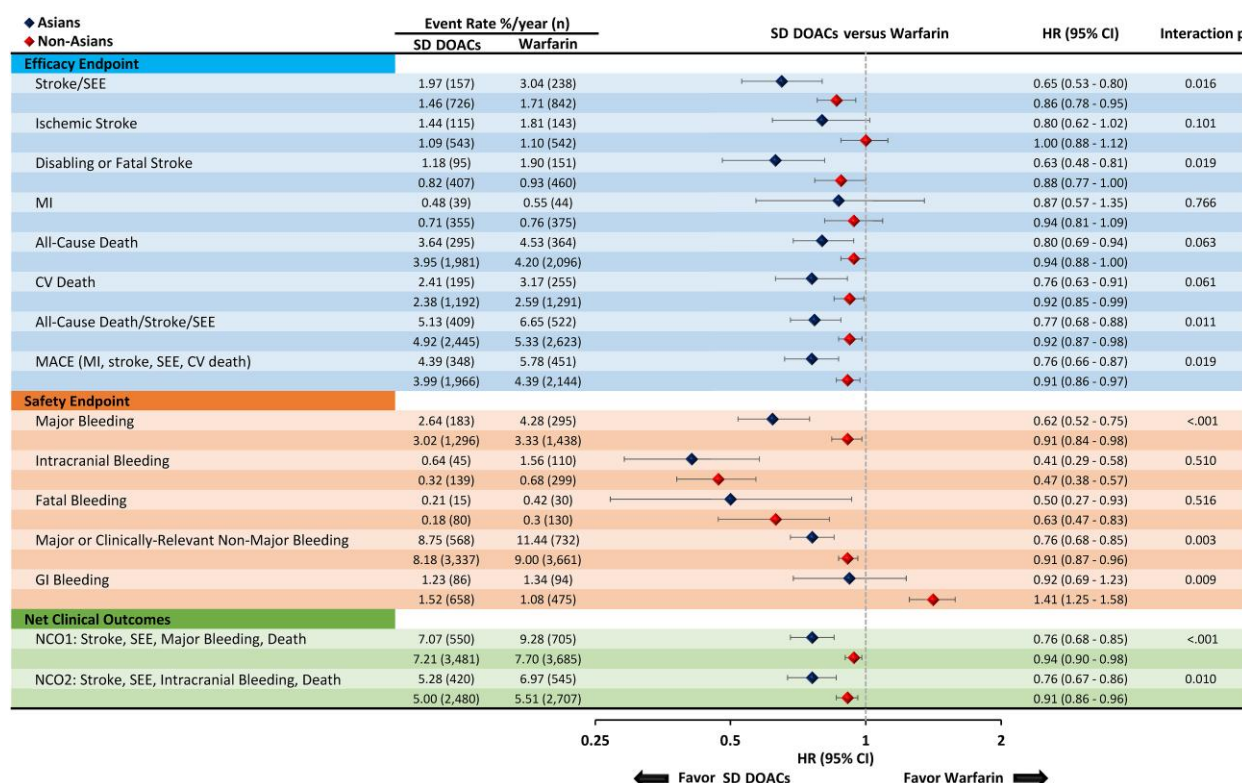


Figure 2 Risks of clinical events (standard-dose direct oral anticoagulants vs warfarin) in Asians and non-Asians. The benefits of standard-dose direct oral anticoagulants over warfarin in reducing risks of ischaemic events, major bleeding, and net clinical outcomes were significantly greater in Asians than non-Asians. The risk of gastrointestinal bleeding was higher with standard-dose direct oral anticoagulants compared with warfarin among non-Asians, while the risk was similar between standard-dose direct oral anticoagulants and warfarin in Asians. CI, confidence interval; CV, cardiovascular; GI, gastrointestinal; HR, hazard ratio; MACE, major adverse cardiovascular event; MI, myocardial infarction; NCO, net clinical outcome; SD DOAC, standard-dose direct oral anticoagulant; SEE, systemic embolic event

Asians, there was no evidence of effect modification by BW for SD DOACs compared with warfarin regarding the risks of stroke/SEE ($P_{\text{int}} = .38$), MB ($P_{\text{int}} = .13$), ICH ($P_{\text{int}} = .48$), and NCO1 ($P_{\text{int}} = .14$) (Figure 3). Similarly, across the spectrum of renal function, SD DOACs also reduced the risks of clinical events in Asians ($P_{\text{int}} = .66$ for stroke/SEE, $P_{\text{int}} = .63$ for MB, $P_{\text{int}} = .73$ for ICH, and $P_{\text{int}} = .46$ for NCO1) (Figure 4).

Comparison of lower-dose direct oral anticoagulants vs warfarin in Asians vs non-Asians

The safety and efficacy profiles of LD DOACs compared with warfarin stratified by race are shown in Supplementary data online, Figure S10. There was no evidence of significant modification of the treatment effect by race for most efficacy, safety, or NCOs for the comparisons of LD DOAC vs warfarin (each $P_{\text{int}} > .05$). Two important exceptions were MB (HR in Asians .45 [.34-.60] vs .66 [.59-.74] in non-Asians; $P_{\text{int}} = .018$) and major or CRNM bleeding (HRs .56 [.48-.66] vs .68 [.63-.73]; $P_{\text{int}} = .044$), both of which were reduced to an even greater degree in Asians than non-Asians treated with LD DOACs vs warfarin.

Comparisons of lower-dose direct oral anticoagulants vs standard-dose direct oral anticoagulants

Lower-dose DOACs compared with SD DOACs increased the risk of stroke/SEE and ischaemic stroke and reduced the risk of MB for Asians and non-Asians to a similar degree ($P_{\text{int}} > .05$ for each) (see Supplementary data online, Figure S11). There were weak statistical trends suggesting more favourable outcomes with SD DOACs for the NCOs in Asians relative to non-Asians ($P_{\text{int}} = .076$ for NCO1 and $P_{\text{int}} = .10$ for NCO2).

Specifically in Asians, LD DOACs had a 57% higher risk of stroke/SEE (HR 1.57, 95% CI 1.15-2.13, $P = .004$) but a 30% lower risk of MB (HR .70, 95% CI .51-.96, $P = .026$) compared with SD DOACs (Figure 5). The higher risk of stroke/SEE with LD DOAC vs SD DOAC was driven by a near doubling in the risk of ischaemic stroke with LD DOACs in Asians compared with SD DOACs (HR 1.92 [1.35-2.73], $P < .001$). The risk of NCO1 was similar, while the risk of NCO2 was 23% higher for LD DOACs vs SD DOACs (HR 1.23 [1.03-1.48], $P = .025$). There was no significant heterogeneity between the two LD DOACs (dabigatran 110 mg twice daily and edoxaban 30 mg/dose reduced to 15 mg) in relation to the findings mentioned above (all $P_{\text{int}} > .05$).

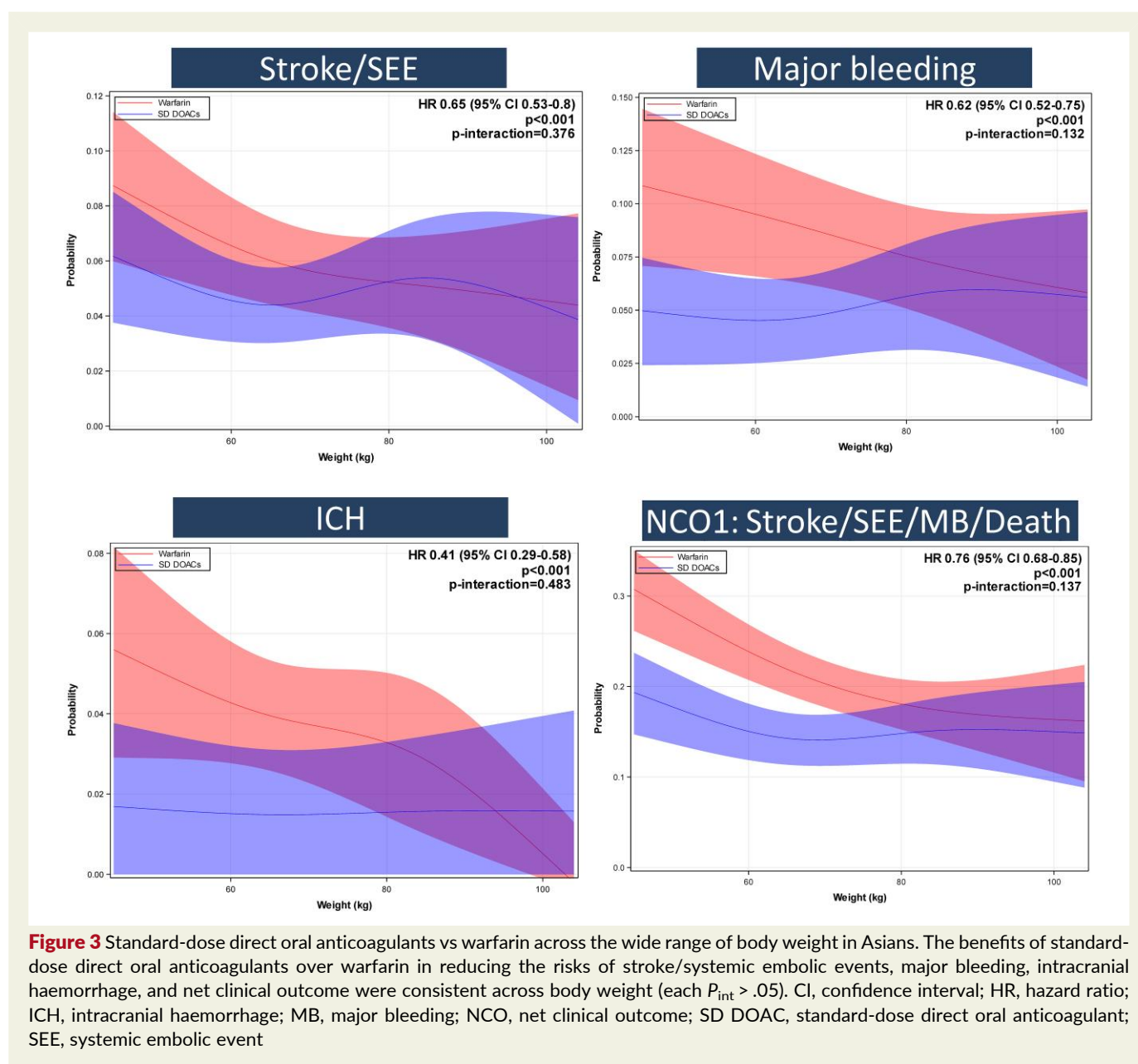


Figure 3 Standard-dose direct oral anticoagulants vs warfarin across the wide range of body weight in Asians. The benefits of standard-dose direct oral anticoagulants over warfarin in reducing the risks of stroke/systemic embolic events, major bleeding, intracranial haemorrhage, and net clinical outcome were consistent across body weight (each $P_{\text{int}} > .05$). CI, confidence interval; HR, hazard ratio; ICH, intracranial haemorrhage; MB, major bleeding; NCO, net clinical outcome; SD DOAC, standard-dose direct oral anticoagulant; SEE, systemic embolic event

Discussion

In this analysis from the COMBINE AF database, our principal findings are as follows: (i) patients with AF of Asian race differed from non-Asians in clinical characteristics—most notably Asians on average were 19.5 kg lighter in BW with a 12.4 mL/min lower CrCl; (ii) compared with non-Asian patients, Asians treated with warfarin had a significantly higher adjusted risk of several clinical outcomes, including both efficacy (stroke/SEE, ischaemic stroke, disabling or fatal stroke) and safety (MB, ICH, GI bleeding) endpoints; (iii) the benefits of SD DOACs over warfarin in reducing risks of ischaemic events, bleeding events, and NCOs were significantly greater in Asians than non-Asians; (iv) in Asians, the advantages of SD DOACs over warfarin were consistent across the wide range of BW and CrCl; and (v) SD DOACs had a lower risk of stroke/SEE (in

particular ischaemic stroke) and a higher risk of MB compared with LD DOACs for both Asians and non-Asians. Specifically, in Asians, NCOs tended to favour SD DOACs over LD DOACs (*Structured Graphical Abstract*). To our knowledge, this is the first individual patient-level meta-analysis of the pivotal DOAC trials directly comparing clinical outcomes between Asians and non-Asians, with additional stratification by TTR, as well as a comprehensive evaluation of SD DOACs vs warfarin in clinically relevant subgroups and across the full ranges of BW and CrCl in Asians.

Risks of stroke and bleeding for Asians vs non-Asians randomized to warfarin

In the present study, we demonstrated significantly higher adjusted risks of stroke and bleeding events in Asians compared with non-Asians treated with warfarin. Although the median

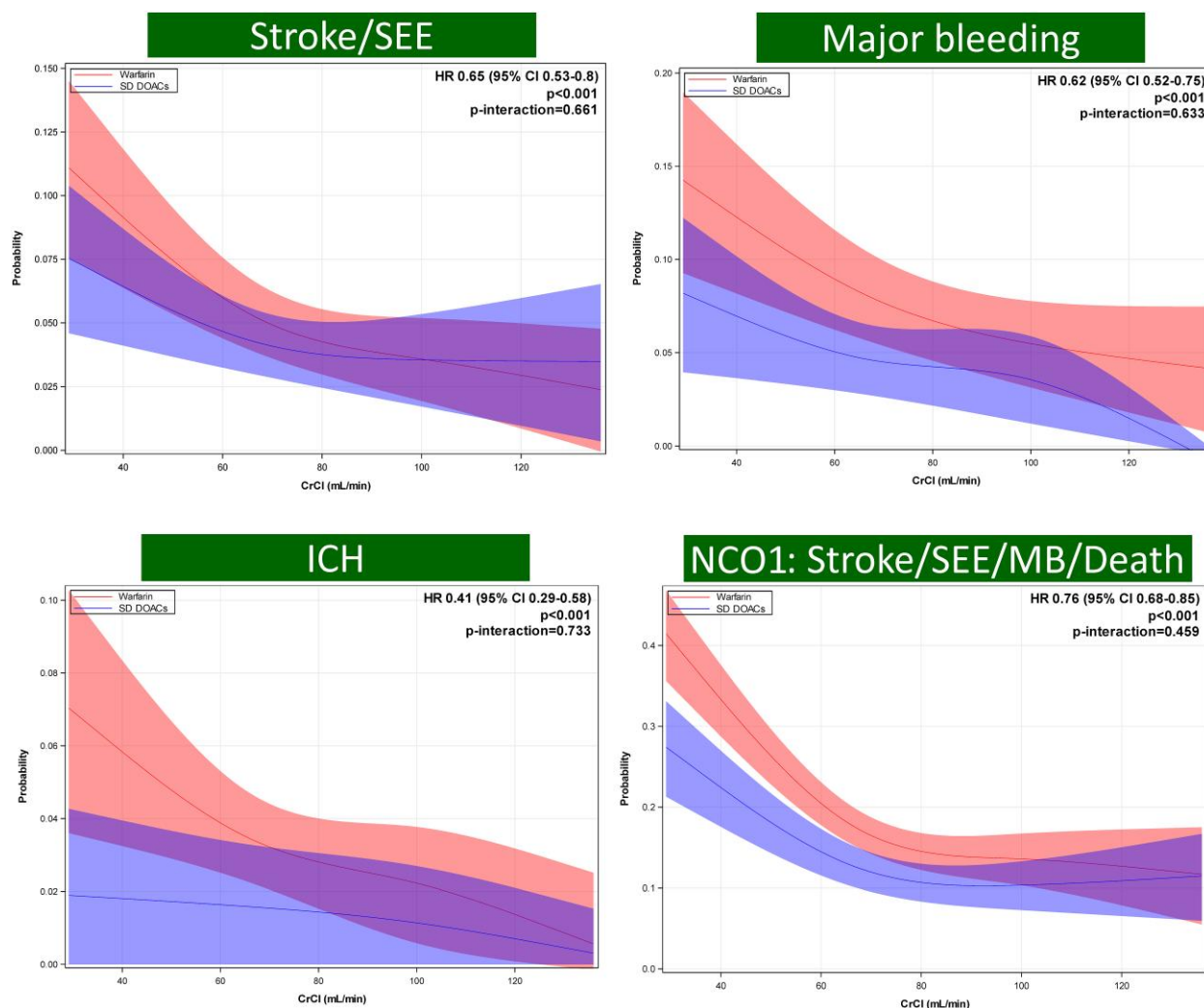


Figure 4 Standard-dose direct oral anticoagulants vs warfarin across the range of creatinine clearance in Asians. The benefits of standard-dose direct oral anticoagulants over warfarin in reducing the risks of stroke/systemic embolic events, major bleeding, intracranial haemorrhage, and net clinical outcome were consistent across creatinine clearance (all $P_{\text{int}} > .05$). CI, confidence interval; CrCl, creatinine clearance; HR, hazard ratio; ICH, intracranial haemorrhage; MB, major bleeding; NCO, net clinical outcome; SD DOAC, standard-dose direct oral anticoagulant; SEE, systemic embolic event

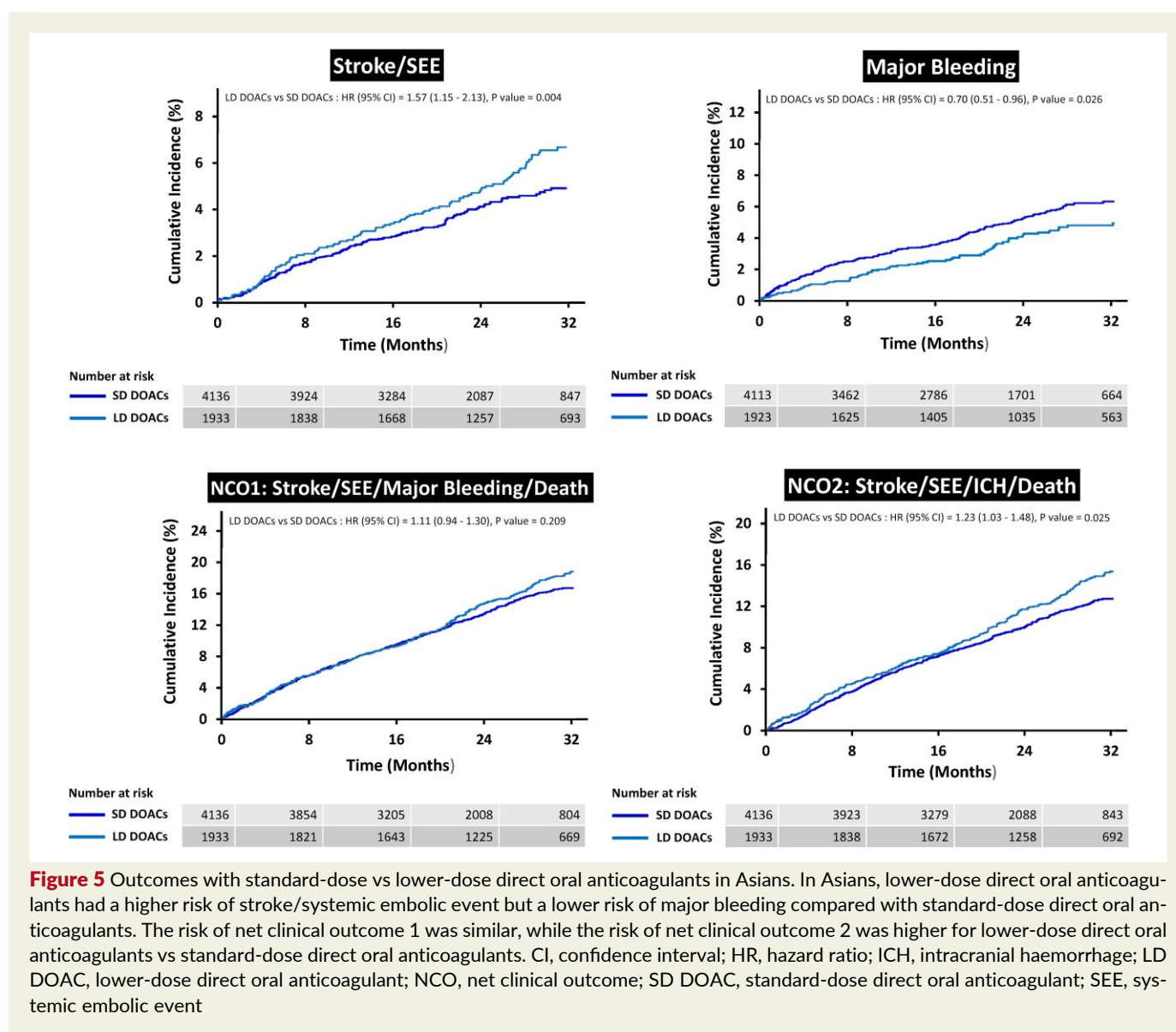
TTR was 8.5% lower in Asians (which is associated with worse outcomes with vitamin K antagonists²⁶), elevated adjusted risks for ischaemic and bleeding events, as well as net outcomes in Asians vs Non-Asians, were still noted in patients with a TTR > 70%. These findings suggest that Asians represent a high-risk cohort for whom anticoagulants other than VKAs should be considered, even if a high TTR could be achieved.

Advantages of standard-dose direct oral anticoagulants in Asians over warfarin

In COMBINE AF, SD DOACs significantly decreased the risks of stroke/SEE by 19%, all-cause mortality by 8%, and ICH by 55% compared with warfarin.¹⁴ In our analysis of Asian race, SD DOACs reduced stroke/SEE and disabling or fatal stroke to a greater degree in Asians than non-Asians compared with warfarin. The larger benefits of DOACs over warfarin in Asians

may be partly explained by the higher risk of ischaemic events in the warfarin arm, possibly related to the lower TTR observed in Asians than in non-Asians. Of note, although more Asian AF patients met the dosage reduction criteria of DOACs and received the reduced-dose DOACs per protocol, the efficacy of DOACs was still maintained. Specifically, in a subgroup analysis of patients with a $\text{CrCl} \leq 50$ mL/min or $\text{BW} \leq 60$ kg, greater reductions in stroke/SEE with FXai vs warfarin were observed in Asians compared with non-Asians.

When compared with warfarin, DOACs also reduced the risk of MB to a larger degree in Asians than non-Asians. This observation may be partly explained by the greater percentage of Asian who qualified for a reduction in DOAC dose and is supported by the greater reduction in MB with FXai in Asians with a $\text{CrCl} \leq 50$ mL/min or $\text{BW} \leq 60$ kg compared with non-Asians. Interestingly, the risk of GI bleeding was higher with DOACs vs warfarin only in non-Asians (HR 1.41), but not



in Asians (HR .92) ($P_{\text{int}} = .009$). Data from ENGAGE AF-TIMI 48 found that a higher rate of dose reduction in Asians contributed to lower edoxaban drug levels and inhibition of endogenous FXa, thus mitigating the increased risk of GI bleeding seen in non-Asians.^{11,27} Additional studies are necessary to confirm this finding.

Standard-dose direct oral anticoagulants vs warfarin across the range of body weight and creatinine clearance in Asians

In prior secondary analyses of randomized trials, the efficacy and safety of DOACs vs warfarin were generally consistent in patients across the spectrum of BW^{24,25} and CrCl categories,^{20,22,23,28} with the possible exception of patients at the very high end of BW (>100 kg)¹⁶ or CrCl (>110 mL/min).¹⁵ However, similar analyses, specifically in Asians, are limited. This issue is particularly important for the Asian population since they have, on average, a substantially lower BW and CrCl

compared with non-Asians. These observations have generated clinical concerns regarding the safety of DOACs (particularly MB and ICH) in Asians with low BW and CrCl. Notably, the European Heart Rhythm Association (EHRA) practical guide on the use of DOACs in AF recommends the measurements of plasma concentrations for patients with a BW < 60 kg and even considering warfarin for those with a BW < 50 kg.²⁹ However, these recommendations are largely based on expert opinion and should therefore be interpreted with caution. Subsequent data from the ENGAGE AF-TIMI 48 trial demonstrated that the pharmacokinetic/pharmacodynamic profile of edoxaban was consistent across extreme BW groups (<5th percentile, ≤55 kg; 45th–55th percentile, 79.8–84 kg; and >95th percentile, ≥120 kg), and net outcomes were most favourable with edoxaban as compared with warfarin in low-BW patients.^{25,30} Different from EHRA, the AF guidelines of the Asia Pacific Heart Rhythm Society did not comment on this issue due to a lack of data in Asian patients.¹⁸ In the present analysis, we showed that SD DOACs reduced risks of ischaemic and bleeding events

consistently across a wide range of BW (down to 40 kg) and CrCl (down to 30 mL/min) compared with warfarin ($P_{\text{int}} > .10$ for each endpoint) in Asians. Therefore, these data provide reassurance for clinicians that SD DOACs are safe and effective in patients of Asian race, even at lower BWs or CrCl (down to 30 mL/min).

Standard-dose direct oral anticoagulants or lower-dose direct oral anticoagulants for Asians?

There have been discussions about whether LD DOACs could be an alternative for Asians given their observed higher rates of bleeding. Initial analyses in Asians were limited due to their observational design^{31,32} or lack of a comparator.³³ A prior sub-analysis of ENGAGE AF-TIMI 48 trial comparing clinical outcomes with higher (60/30 mg per day) vs LD edoxaban (30/15 mg per day) regimens in Asians and non-Asians supported the use of SD edoxaban in Asians, but included only a modest number ($n = 1942$) of Asian patients and evaluated a single DOAC.^{11,34} The larger sample size of COMBINE AF dataset with >10 000 patients of Asian race treated with one of four DOACs vs warfarin represents a strength of the current analysis. Standard-dose DOACs had a lower risk of stroke/SEE but a higher risk of MB compared with LD DOACs for both Asians and non-Asians. However, net outcomes in Asians with SD DOAC vs LD DOAC tended to be more favourable in Asians compared with non-Asians. These data are important because underdosing of DOACs is not uncommon in Asia. In the ETNA-AF registry, inappropriate underdosing of edoxaban at 30 mg/day accounted for 12.6% and 19.1% of all edoxaban prescriptions in Japan and South Korea/Taiwan, respectively, which are both higher than the rate observed in Europe (8.5%).³⁵ In addition, the ELDERCARE regimen (edoxaban 15 mg/day)³⁶ has been approved for stroke prevention in AF in several Asian countries (Japan, South Korea, Taiwan, the Philippines, etc.). We are concerned that this may further result in the widespread use of underdosed DOAC regimens. Our data support that SD DOACs (with dose reduction as per the label) should remain the preferred initial choice over off-label use of LD DOACs for Asian AF patients. It should also be emphasized that the edoxaban 30 mg/15 mg regimen has not been approved by the US Food and Drug Administration or the European Medicines Agency for stroke prevention in AF.

Study limitations

There are several limitations of the present study. First, patients in the COMBINE AF database are from randomized trials with specific inclusion and exclusion criteria. Therefore, our results may not be generalizable to an unselected group of patients with AF encountered in clinical practice. Second, we took a pragmatic approach by using BW < 60 kg or CrCl < 50 mL/min as the criteria for dose reduction of DOACs, which most closely approximated dose reduction across the trials. However, the criteria for dose reduction varied across the three factor Xa inhibitor trials and were not used with dabigatran in RE-LY. Third, despite the large number of patients enrolled in these four pivotal trials, some subgroup analyses, particularly for less frequent events, had limited power. Despite multivariable adjustment in the comparisons between Asians and non-Asians, we cannot exclude the possibility of residual confounding due

to unmeasured factors, selection bias in enrolment, or differences of healthcare systems related to race. We acknowledge that only two trials studied a LD DOAC arm; however, no other large, randomized datasets of LD regimens in AF exist. Lastly, although more than 10 000 Asians and 60 000 non-Asians were included, the trials and this meta-analysis were not powered to detect differences across multiple endpoints between races. Therefore, these results should be considered exploratory, as we did not control for Type I error nor adjust for multiple comparisons.

Conclusions

Use of warfarin in patients with AF of Asian race was associated with higher adjusted risks of stroke/SEE and MB compared with warfarin treatment in non-Asians. Although both Asians and non-Asians benefit from DOACs over warfarin, the reductions in the risks of stroke/SEE, MB, and NCOs associated with DOAC use were significantly greater in Asians than non-Asians. Even in Asian patients with low BW and moderate renal dysfunction, the advantages of SD DOACs compared with warfarin were consistent. Standard-dose DOACs should remain the preferred initial choice for stroke prevention in patients of Asian race with AF.

Supplementary data

Supplementary data are available at [European Heart Journal](#) online.

Declarations

Disclosure of Interest

E.B. has received consulting fees from Amgen, Boehringer Ingelheim/Lilly, Cardurion, MyoKardia, Novo Nordisk, and Verve. J.W.E. reports honoraria and grant support from AstraZeneca, Bayer, Boehringer Ingelheim, Bristol Myers Squibb/Pfizer, Daiichi Sankyo, GlaxoSmithKline, Janssen, sanofi-aventis, and Eli Lilly as well as a personnel award from the Heart and Stroke Foundation. C.B.G. reports personal fees from Bayer and Boston Scientific; grants and personal fees from Boehringer Ingelheim, Bristol Myers Squibb, Janssen, and Pfizer; and grants from Daiichi Sankyo during the conduct of the study; personal fees from AbbVie, Espero, Medscape, Medtronic Inc., Merck, the National Institutes of Health, Novo Nordisk, Roche, Rho Pharmaceuticals, CeleCor, Corveio, Philips, Abiomed, and Anthos Therapeutics; grants from Akros, AstraZeneca, the US Food and Drug Administration, GlaxoSmithKline, Medtronic Foundation, and Apple; and grants and personal fees from Novartis and The Medicines Company outside the submitted work. S.G. received modest amount of personal fee from Amgen, Merck Sharp & Dohme (MSD), Jansen Pharma, and Anthos Therapeutics. S.G. received quality fee from the American Heart Association as an Associate Editor for Circulation and Duke University and Harvard University as a Steering Committee Member for Clinical Trials. S.G. received a grant from the Nakatani Foundation for Advancement of Measuring Technologies in Biomedical Engineering, Suzuken Memorial Foundation, and the Cross-ministerial Strategic Innovation Promotion Program (SIP) on 'Integrated Health

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Data Availability

The data underlying this article are available in the article and in its online [supplementary material](#).

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Ethical Approval

All patients provided written informed consent, and approval from local ethics committees had been obtained prior to initiation of the RE-LY, ROCKET AF, ARISTOTLE, and ENGAGE AF-TIMI 48 trials.

Pre-registered Clinical Trial Number

Not applicable.

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